

Q1 2024 Earnings Call

Company Participants

- Christophe Fouquet, Chief Business Officer
- Peter Wennink, President and Chief Executive Officer
- Roger Dassen, Executive Vice President and Chief Financial Officer
- Skip Miller, Head of Investor Relations

Other Participants

- Alexander Duval, Analyst, Goldman Sachs
- C.J. Muse, Analyst, Cantor Fitzgerald
- Chris Caso, Analyst, Wolfe Research
- Didier Scemama, Analyst, Bank of America
- Francois-Xavier Bouvignies, Analyst, UBS
- Joe Quatrochi, Analyst, Wells Fargo
- Krish Sankar, Analyst, TD Cowen
- Mehdi Hosseini, Analyst, Susquehanna
- Tammy Qiu, Analyst, Berenberg Bank

Presentation

Operator

Good day and thank you for standing by. Welcome to the ASML 2024 First Quarter Financial Results Conference Call on April 17, 2024. At this time, all participants are in a listen only mode. After the speakers' introduction, there'll be a question-and-answer session. (Operator Instructions) Please be advised that today's conference is being recorded.

I would now like to hand the conference call over to Mr. Skip Miller. Please go ahead.

Skip Miller {BIO 20244900 <GO>}

Thank you, operator. Welcome, everyone. This is Skip Miller, Vice President of Investor Relations at ASML. Joining me today on the call are ASML's CEO, Peter Wennink; our CFO, Roger Dassen; and our Chief Business Officer and Incoming CEO, Christophe Fouquet.

The subject of today's call is ASML's 2024 first quarter results. The length of this call will be 60 minutes and questions will be taken in the order that they are received. This call is also being broadcast live over the Internet at asml.com. A transcript of management's opening remarks and a replay of the call will be available on our website shortly following the conclusion of this call.

Before we begin, I'd like to caution listeners that comments made by management during this conference call will include forward-looking statements within the meaning of the Federal Securities laws. These forward-looking statements involve material risks and uncertainties. For a discussion of risk factors, I encourage you to review the Safe Harbor statement contained in today's press release and presentation found on our website at asml.com and in ASML's Annual Report on Form 20-F and other documents as filed with the Securities and Exchange Commission.

With that, I'd like to turn the call over to Peter Wennink for a brief introduction.

Peter Wennink {BIO 1852674 <GO>}

Thank you, Skip. Welcome, everyone, and thank you for joining us for our first quarter 2024 results conference call. Before we begin the Q&A session, Roger, Christophe and I would like to provide an overview and some commentary on the first quarter 2024 as well as provide our view of the coming quarters and Roger will start with a review of our first quarter 2024 financial performance with added comments on our short-term outlook and I will briefly reflect on the current market environment and then hand over to Christophe to complete the introduction with some additional comments on the future business outlook. Roger?

Roger Dassen {BIO 15064806 <GO>}

Thank you, Peter, and welcome everyone. I will first review the first quarter 2024 financial accomplishments and then provide guidance on the second quarter of 2024. Let me start with our first quarter accomplishments. Total net sales come in at EUR5.3 billion at the midpoint of guidance. We shipped 12 EUV systems and recognized EUR1.8 billion revenue from eleven systems this quarter.

Net sales -- net system sales of EUR4 billion which was driven by logic at 63% with the remaining 37% coming from memory. Installed Base Management sales for the quarter came in at EUR1.3 billion as guided. Gross margin for the quarter came in at 51% which is above our guidance, primarily driven by product mix or immersion in UV systems and someone else. On operating expenses, R&D expenses came in at EUR1.32 billion and SG&A expenses came in at EUR273 million, both slightly lower than guided due to a shift in spend to later in the year. Net income in Q1 was EUR1.2 billion, representing 23.1% of total net sales and resulting in an EPS of EUR3.11.

Turning to the balance sheet, we ended the first quarter with cash, cash equivalents and short-term investments at a level of EUR5.4 billion, which is lower than previous quarter. We ended Q1 with negative free cash flow, primarily driven by lower down payments and higher inventory relative to last quarter. In the current environment, as customers work to return to profitability and strengthen cash position, we continue to provide some support for our customers. The higher inventory is a result of the increased material intake, including High NA as part of planned capacity ramp in preparation for stronger demand next year.

Moving to the order book, Q1 net system bookings came in at EUR3.6 billion, which is made up of EUR656 million for EUV bookings and EUR2.9 billion for non-EUV bookings. Net system bookings in the quarter were driven by memory at 59% and logic for the remaining 41% of the bookings. There is quite a bit of speculation around order numbers, so I will make a few comments here. In the past six months, we've had orders of almost EUR13 billion, which is quite significant. As we said in the past, our order flow can be lumpy and may not be evenly distributed over the year. Although we don't guide orders, an order rate, a bit over EUR4 billion per quarter for the final three quarters of the year, would provide full order coverage at the end of 2024 for a 2025 sales number that would be at the midpoint of our 2022 Investor Day scenarios. At the end of Q1 2024, we finished with a backlog of around EUR38 billion.

With that, I would like to turn to our expectations for the second quarter of 2024. We expect Q2 total net sales to be between EUR5.7 billion and EUR6.2 billion. We expect our

Q2 Installed Base Management sales to be around EUR1.4 billion. The relatively low first half of the year compared to the expected strong second half is in line with the expected industry recovery from the downturn. Gross margin for Q2 is expected to be between 50% and 51%. The expected R&D expenses for Q2 are around EUR1.70 billion and SG&A is expected to be around EUR295 million.

Our estimated 2024 annualized effective tax rate, this leads to a final dividend proposal to the annual general meeting of EUR1.75 per ordinary share, which will result in a total dividend for the year 2023 of EUR6.10 per ordinary share, which is a 5.2% increase compared to 2022. In Q1 2024, we purchased around 0.5 million shares for a total amount of around EUR400 million.

With that, I would like to turn the call back over to Peter.

Peter Wennink {[BIO 1852674 <GO>](#)}

Thank you, Roger. As Roger has highlighted a relatively slow Q1 as start to the year is consistent with our guidance and expectations coming out of the downturn. Overall semiconductor industry -- sorry, overall semiconductor inventory levels continue to improve, trending towards more healthy levels. We also see continued improvements in lithography tool utilization at both logic and memory customers, all in line with the industry's continued recovery from the downturn.

Looking at the market segments, we see a similar environment as communicated last quarter with demand momentum from AI-related applications. Memory demand is primarily driven by DRAM technology node transitions in support of advanced memories such as DDR5 and HBM. Logic customers continue to digest the significant capacity additions made over the last year -- over the past year. As many of you know, next week, April 24, is the general meeting of shareholders and my last effective working day at ASML. Although this is not a big surprise anymore, it's still a big event for me, Martin, and our families and it has been an enormous privilege of being able to serve the company and its many stakeholders for so long. I have thoroughly enjoyed virtually every moment of it and the many interactions I've had with many of you, including these conference calls, believe it or not, and I hope I will see some of you someday, sometime and wish you all good health, a prosperous and happy life.

And with that, I'd like to turn over to you, Christophe.

Christophe Fouquet {[BIO 20524437 <GO>](#)}

Thank you, Peter. And first of all, thank you for the last 10 years leading ASML and making it the great company we know today. I think some of our audience have been with you for the 40 plus quarters you led as CEO, but probably not many for the nearly 100 quarterly calls over your past 25 years in ASML. I am sure everyone on the call will miss you as much as we all will at ASML. I am myself very honored and privileged to succeed Peter and I am very much looking forward to working with all of you.

As Peter mentioned, our view on the market segment for 2024 has not changed relative to what we stated last quarter. We expect memory revenue growth this year, primarily driven by technology transition in support of advanced memory technology. We see lower logic revenue this year relative to last year as customers digest litho capacity installed over the past year.

Turning to our businesses. For EUV, we continue to expect revenue growth in 2024. We plan to recognize revenue on a similar number of EUV 0.33 NA system as 2023. In addition, we expect revenue from one to two High NA systems. On our 0.33 NA system, we shipped the first NXE:3800E this quarter for qualification at the customer. The NXE:3800E has the capability to deliver a significant increase in performance with a productivity of 220 wafers per hour, which is a 37% increase over the NXE:3600D in its final configuration.

The NXE:3800E also brings imaging and overlay improvements, which will make it the future tool of choice for memory and logic advanced nodes. Those performance increases will deliver better value for our customers, including cost of ownership, and will translate into higher ASPs and improve margins for ASML. EUV customer plan to transition to the NXE:3800E this year. As a result, the majority of our low NA EUV shipments in the second half of the year will be this system.

Regarding High NA or 0.55 NA EUV, we shipped our first system to a customer and this system is currently under installation. We started to ship the second system this month and its installation is also about to start. During the SPIE Industry Conference in February, we announced first light on our High NA system located in our joint ASML-imec High NA in Veldhoven. We have since achieved first images with a new record resolution below 10 nanometer and expect to start exposing wafers in the coming weeks. All High NA customers will use this system for early access to process development.

The customer interest for our system lab is high as this system will help both our logic and memory customers prepare for High NA insertion into their roadmaps. Relative to 0.33 NA, the 0.55 NA system provide finer resolution enabling an almost three times increase in transistor density at a similar productivity in support of sub 2 nanometer logic and sub 10 nanometer DRAM nodes. We expect our non-EUV business to be down in 2024, primarily driven by lower emerging system sales relative to 2023.

For our Installed Base business, based on our view today, we expect a similar level of revenue compared to last year. As the recovery becomes more clear this year, customer may look to upgrade their system in preparation for 2025 and this could provide future business opportunity this year. Our outlook for the full year is unchanged with similar revenue compared to 2023. In line with the industry continued recovery from the downturn, we expect a stronger second half relative to the first half of the year. We view 2024 as a transition year and continue to make investment this year both in capacity ramp and in technology, to be ready for the upturn in the cycle.

Looking longer term, while there are still significant uncertainties, primarily driven by the macroenvironment, it appears we are passing through the bottom of this specific cycle and we expect an industry recovery over the course of 2024. Based on the discussion with our customers and supporting our strong backlog, we expect 2025 to be a strong year driven by a number of factors as mentioned last quarter.

First, the secular growth driver in semiconductor end markets which we have previously discussed such as energy transition, electrification and AI. The expanding application space, along with increasing lithography on future technology nodes, drive demand for both advanced and material nodes. Second, the industry expects to be in the middle of a cyclical upturn in 2025. And last, as mentioned earlier, we need to prepare for the significant number of new fabs that are being built across the globe, in some instances clearly supported by several government incentive plans. These fabs are spread geographically, are strategic for our customers and are scheduled to take our tool.

It is essential that we keep our focus on the future and build capacity in preparation for further long-term growth as we discussed in the market scenarios for 2025 and 2030 during our Investor Day in November 2022. We plan to update our view during our Investor Day this year on November 14, 2024. In summary, also there is near-term uncertainty. We remain confident in our long-term growth opportunity.

With that, we will be happy to take your questions.

Skip Miller {BIO 20244900 <GO>}

Thank you, Roger, Peter and Christophe. The operator will instruct you momentarily on the protocol for the Q&A session. Beforehand, I would like to ask you if you kindly limit yourself to one question with one short follow up if necessary. This will allow us to get to as many callers as possible. Also, as the CEO transition is planned next week following the AGM on April 24th, Roger and Christophe will take the majority of the questions as it pertains to the forward-looking comments.

Now, operator, could we take your final instructions and then the first question please?

Questions And Answers

Operator

Thank you. (Operator Instructions) We will now go to your first question. And your first question comes from the line of Krish Sankar from TD Cowen. Please go ahead.

Q - Krish Sankar {BIO 16151788 <GO>}

Hi, thanks for taking my question. And Peter and I guess also for Martin as well, thanks for everything over the years. You both will be missed for sure. And then I guess my first question is for Roger. I understand bookings can be lumpy, but the EUV orders are also down quite a lot in the March quarter. And you said that you need to hit over EUR4 billion run rate to hit the midpoint of calendar '25. But I'm just kind of curious, there's an expectation that it should be better than the midpoint of next in -- outlook for next year. Do we really need to see a pickup in EUV orders to really meet those calendar '25 outlooks? And where do you think that's going to come from mainly? Is it the foundry-logic vertical? And also just along the same path, how much of your memory bookings was from China? And then I had a follow up.

A - Roger Dassen {BIO 15064806 <GO>}

And a follow up. Many questions in one question, Krish. Very well done. I'll try and answer them -- I'll try and answer them as best as I can. But yeah, Krish, I think you're absolutely right and our conclusion that indeed order intake is very lumpy and I think that's what we've seen in the past and we have been saying that for many, many years. We said it last time when the orders came in very high. We say today when from the vantage point of some, the orders come in pretty low. And therefore, if you look at the past six months combined, you're looking at EUR13 billion, which is EUR6.5 billion per quarter, which we still continue. We still believe it's pretty significant. When I talk about the EUR4 billion that we need in order to get to the midpoint of next year, I'm indeed talking about everything. So, of course, that also includes EUV.

And I think, as many of you have probably recognized, if you look at the intake in the past couple of quarters and also in the past quarter, it's pretty clear that there's a few usual suspects absent in the order intake. And I think that's pretty clear, right? So if we look at the plans of some of our large customers and you talk about foundry, foundry indeed does come to mind in this discussion. If you look at the plans and the announced plans of some of our larger customers, it's pretty clear that in the next couple of quarters, significant orders need to come in. So -- and part of the EUR4 billion that you should see in order to get to this midpoint, indeed have to include the orders from those customers.

So again, the midpoint, does that mean that now all of a sudden, we're guiding midpoint? No, we're not. As Peter has very clearly said it on previous calls, if we look at 2025, we're looking at a significant uptick. We're not looking at the low end of the scenarios that we provided back in 2022, but we're not saying now you need to look at the midpoint of the guidance. We just give you the math that is required in order to get to the midpoint of the guidance with our expectation of a very strong recovery into 2025 has by no means changed.

In terms of your questions on memory and particularly then on China, of course, as you know, we typically do not disclose the geographic distribution of our order intake. Of course, there is a, I would say, healthy part in the order intake that is related to China, but it's not like -- the order intake is distributed over the globe. So the very high concentration that you saw in the sales for Q1, you don't see that back in the order intake. So the order intake is more distributed geographically than what you would see in the sales for Q1.

Q - Krish Sankar {BIO 16151788 <GO>}

Got it (Multiple Speakers). Yes, thank you very much. And just a very, very quick follow up and really appreciate it. Obviously, with all these like incremental news coming on US and Dutch rules and regulation. Just curious, has that changed your view on what it means for your China sales? Three months ago, it said it's a 10% to 15% impact. Just want to see if there's any updated view on this. Thank you very much.

A - Roger Dassen {BIO 15064806 <GO>}

No, Krish, nothing has changed that. I mean, when we talked about the 10% to 15%, I think that's even longer back when we made that comment. That was directly the consequence of the fact that we realized that this year, we will probably not get licenses to ship the latest generation of inversions [ph]. That was what that 10% to 15% comment was related to. Our perspective has not changed. The rules haven't changed. Of course, there is continued discussion on export controls. The rules haven't changed. Our perspective on the year hasn't changed. We're still looking at a strong sales level for China for this year.

Q - Krish Sankar {BIO 16151788 <GO>}

Thank you.

Operator

Thank you. We will now go to the next question. And your next question comes from the line of Tammy Qiu from Berenberg. Please go ahead.

Q - Tammy Qiu {BIO 17604871 <GO>}

Hi. Thank you for taking my question. So first one is on China, please. Can you talk about your China business trend over the recent quarter, please? Because China has been really strong and there has been always concern that China may actually go to capacity by just impurity in this year or later this year. Do you have any comment on China trend, please? Then I have a follow up.

A - Roger Dassen {BIO 15064806 <GO>}

Yeah, I mean, China, relatively speaking is high, but if you look at absolute numbers, you would recognize that China is actually lower in Q1 than it was in Q4 of last year, right. So - - because if you do the math, then China was at EUR1.9 billion in this quarter. It was at EUR2.2 billion last quarter. So I think from that vantage point, it's gone down a bit, but it's still strong. And the reason that China is strong both in absolute terms and relative terms is also because the rest of the world, the demand is -- or at least the sales and the shipments in Q1 were relatively low, which was no surprise. I think it's very much in sync with our perspective on a market that is in recovery.

And a market being in recovery means that customers are first driving up the utilization of their tools, which is exactly what they're doing and we notice that. And if the utilization comes to a certain point, then they will start this. They will order and they will require shipments. And that's exactly why we've also said on the call that we believe that we're going to see momentum building up in the course of this year with a much stronger second half, as Peter also just had a much stronger second half than the first half.

So in terms of trends in China, strong first quarter for sure, but not a record quarter, but a strong quarter. We expect China to continue to be strong this year. And obviously also to the -- also you have to see that in relation to the rest of the world that we believe is going to recover. We've also said on previous calls and I just want to reiterate that if you look at the demand for China, the demand in China continues to be strong. And that is related, as we said before, to the fact that the demand for mature technology continues to be strong. And I just point that what we said at the Capital Markets Day in 2022. Capital Markets Day in 2022, we said we believe every single year between '22 and 2030, we believe 380k wafer starts and capacity needs to be added to mature.

And if we look at what has happened last year, what has been added in terms of that both in China and in the rest of the world, it's actually below that number. So yes, China is strong. And, of course, it's not just mature, but mature is a very significant part of what China is added -- is adding. Yes, China is strong, but China is strong because they're adding capacity that we believe the world needs. And yes, as a result of this, China's share and global market share will over the years become larger than it is today. Their self-sufficiency will increase in comparison to today. But we believe that what China is adding today in terms of mature capacity is rational and is in line with the -- with our expectation of what capacity and mature needs to be added in order to get to what the world needs in the second half of the decade.

Q - Tammy Qiu {BIO 17604871 <GO>}

Okay, thank you, Roger. And I have a follow up on the EUV and 2 nanometer, 3 nanometer area, please. So you are expecting some massive order from foundry and logic customers. Can you share is that for 3 nanometer or 2 nanometer? And what is the level

of EUV kind of layer count between those two? Because I do understand that potentially you can have some usage, so that may impair some 2 nanometer incremental demand because some 3 nanometer can be migrated or reused for 2.

A - Christophe Fouquet {BIO 20524437 <GO>}

Yeah, so this is Christophe here. So maybe on your first question, so I think we mentioned a few times now that customer were still logic customer. Foundry customers were still digesting, so some of the capacity they had put in place. We were then referring to 3 nanometer and 5 nanometer. When we look for the 2 nanometer capacity still have to come and I think, as you most probably are aware of, we expect the ramp for that technology to start sometime next year. So I think this will be the next most probably wave of EUV order and this is also back to the comment Roger made in answer to the first question. So we are going to now focus when it comes to EUV and logic foundry, mostly to 2 nanometer of order intake, which, as Roger said, should come in the next few months.

On the number of layers, so no change there. I think we mentioned in the past that the layer -- the EUV layer for 2 nanometer is very similar to what we had on 3 nanometer. Two nanometer is mostly a device transition. As you know, most customers, logic -- foundry customers will transition to (inaudible), which is, I would say, quite a complex move. And as a result, the focus of the change is on that. So all the expectations we have in terms of EUV layer on 2 nanometers are not different from the one we have shared with you for quite a few months already now.

Q - Tammy Qiu {BIO 17604871 <GO>}

Okay. Thank you, Christophe. Peter, happy retirement, and thanks for being with us over the past 10 years.

A - Peter Wennink {BIO 1852674 <GO>}

You're welcome. Thank you.

Operator

Thank you. We will now take the next question. And your next question comes from the line of Joe Quatrochi from Wells Fargo. Please go ahead.

Q - Joe Quatrochi {BIO 18961101 <GO>}

Yeah, thanks for taking the question. I was curious, as we think about the order book and we think about filling out 2025, the company has been very specific about pre-building low and a tools over the course of this year and into next year. Has that changed the way that your customers are thinking about their order cadence?

A - Roger Dassen {BIO 15064806 <GO>}

Well, I mean, that would be a bit opportunistic on their side. And I like to think of the relationship that we have with our customers as a -- as much more one-off partnership than one-off transactional behavior. So, I don't think that necessarily has an impact. Of course, we said in previous calls, in order to get as many -- create as many degrees of flexibility that we have for next year that we will do some pre-building. But, of course, we do that in very close interaction with customers, understanding what they need.

And I think that's a big distinction that I think you need to draw. I mean, on the one hand, we're having very intense interactions with customers to understand what they need. And then you have PO. And actually the PO process, as you probably will appreciate given the amounts that we're talking about these days, are pretty bureaucratic and formal processes where there is a lot of governance necessary in order to get there. That gets you to the lumpiness. But in the meantime, we have a pretty good understanding based on our interactions with customers, what they really need.

So that's I think -- it's the interaction with the customers, it's the comfort that we get based on those conversations that ultimately drives our plans for the year, that ultimately drives our plans for pre-building much more so than whether or not an order is going to be received or not.

A - Peter Wennink {BIO 1852674 <GO>}

Yeah. And perhaps speaking from experience over the last 25 years, we're a transparent company. So it also means that we will build inventory or work (inaudible) to prepare because our lead times are just so long. And we inform you as our shareholders on this. And, of course, customers hear this and then we're in the midst of negotiations on final orders and let's say on a commercial basis, which of course, if you put the two things together, it might be that those orders take a little bit longer, which is quite normal.

So I think this is what we've seen before and we've seen it again. But again, as Roger said earlier, if you believe 2025 and you know that if you want to buy an EUV tool, there's only one phone number that you can actually dial, then this will happen. But this is just where we are in terms of our customers knowing that we're building because we want to make sure that we can ship to them and we have long lead times and their commercial efforts on the POS that will come.

Q - Joe Quatrochi {BIO 18961101 <GO>}

Thanks for that. That's helpful color. As a follow up, just on the memory orders that you're seeing. Just curious is that still more predicated on HBM building out capacity or have you may be seen some green shoots for call it more the conventional DRAM demand?

A - Roger Dassen {BIO 15064806 <GO>}

No, I think the lion's share of the orders that we saw in memory in the last quarter really are still technology related, right? So it's DDR5, it's APM [ph]. That's what most of the orders that we saw are related to that.

Q - Joe Quatrochi {BIO 18961101 <GO>}

Perfect. Thank you.

Operator

Thank you. We will now go to the next question. And your next question comes from the line of Francois Bouvignies from UBS. Please go ahead.

Q - Francois-Xavier Bouvignies {BIO 19672683 <GO>}

Thank you very much. I mean, first of all, Peter, thank you a lot. You will be missed definitely in the investor community and thank you for the dialogue. That was very helpful. And I look forward to work as well more with Christophe. So the first question I had is a bit on the lead times. So you mentioned that the buildup of inventory doesn't impact so much the lead times if I tried to read correctly. And when you said last quarter, you said that you have 12 months or more than 12 months lead time on EUV. Now, when you look at 2025 targets or ranges, you need to have like 70 [ph] more EUV tools to get to this mid to high end. And according to my calculation, I have like around 30 EUV tools in the backlog for next year, which means that you would need more than 40 EUV tools still to satisfy your revenues for 2025.

So I'm just surprised the amount of EUV tools you need. And with the 12 months more lead times, I would think it's a very dangerous game for your customers to wait until the last moment. So can you maybe help us clarify the fact that you probably need orders significantly in a short period of time and reconcile with the fact that you said in the next three quarters you need more than EUR4 billion, but I mean it's going to be too late if it's Q3 or Q4 for EUV to get delivery in '25, if you see what I mean?

A - Christophe Fouquet {BIO 20524437 <GO>}

Yeah, first of all, it's a slightly different question, but essentially the question that we were just looking at, right. It's the same question that Joe was raising to a large extent. And I can just reiterate what I said there. Of course, with customers, we have an ongoing dialogue on what they need. And then Peter said it, then there is a bureaucratic process and there is some negotiation going on that will ultimately lead to the order being made and being translated into a PO that, as you also know these days, comes with the obligation to pay money. And that's probably part of the reason why customers -- why some customers are postponing the placement of the order a little bit as well.

But the reality is that we know quite well what customers want and customers know it as well. So it's just a matter of those two worlds coming together and then ultimately leading to a PO. Is it a dangerous game on their side? Is it a dangerous game on our side? I don't think so. I mean, as long as we have a very open dialogue with one another, I'm pretty sure that in the foreseeable future, you will see the translation of what we know is firm demand into orders. I'm quite confident in that.

A - Peter Wennink {BIO 1852674 <GO>}

And if you believe the 2025 number, which we do, then it's a very high level of mutual dependency here. So that is -- so that's why the game is not that dangerous. We also need each other.

Q - Francois-Xavier Bouvignies {BIO 19672683 <GO>}

Excellent. Thank you. And maybe my follow up is on High NA. I mean, you see -- we saw a couple of milestones in the last few quarters. Christophe, you talked about the productivity, I mean, the density improvements. When we think about the lead times for High NA, what is it today? And if we assume mass production in, let's say, '26, '27, should we expect some uptick in terms of orders for this maybe in the next few quarters? And what's the cumulus tone you are still waiting for from a product point of view, technical point of view, to drive more adoption?

A - Christophe Fouquet {[BIO 20524437 <GO>](#)}

Well, so I think a few key milestones. The first one I think we've been talking about it for a few quarter is the fact that customer have been committing to EUV High NA with double digit basically units in our backlog and they have done that without even seeing one image from the tour. So I think this shows a bit the level of commitment and trust they have in our ability to bring new technology.

Now, I mentioned the first image, 10 nanometer -- less than 10 nanometer resolution image. This is a huge milestone for both our customer and SML because this single image proved that the technology we have been developing for many, many years is working. And you cannot imagine how welcome that milestone was by both our customer and ourselves. That's very, very important.

Now what's going to happen next is since we can soon expose wafer, every customer are going to come to see us here to get access to the tool we have in our lab and start to expose their own vertical, so that they can decide for themselves exactly how to use the tool. And we expect that this works will lead at some point to, I would say, the next set of decision on High NA.

So what's happened in the lab in the next few months, the work we'll be doing with all our customers, and I stress again, all our customer, I think is most probably the most exciting milestone to come and this will really help everyone to understand basically what's next for High NA. But I cannot stress enough on -- I will say how happy and excited that we've been able to generate those first images. When you look at the undertaking of a technology like High NA, for us, for our customer, this is a very, very important achievement. And again, the next few months, we'll build on that.

Q - Francois-Xavier Bouvignies {[BIO 19672683 <GO>](#)}

Thank you very much.

Operator

Thank you. We will now go to the next question. And your next question comes from the line of C.J. Muse from Cantor Fitzgerald. Please go ahead.

Q - C.J. Muse {[BIO 6507553 <GO>](#)}

Yes, thank you for taking my question. And Peter, big congrats. Pleasure working with you. And maybe if I could direct the last question for you here at ASML, given your long experience. As you look at the strong secular trends led by high performance compute, historically at the leading edge, it was led by Apple and Huawei and then only Apple post the embargo. But now given the performance power cost requirement, how are you seeing kind of the trends on the high-performance compute side moving closer and closer to kind of being the pipe cleaner for the bleeding edge? And how does that impact your thinking of what kind of longer-term growth for leading edge workers will look like?

A - Peter Wennink {[BIO 1852674 <GO>](#)}

Well, I think it's two questions, so I answer the second one. You're basically saying, what will drive leading edge high performance compute, but you're absolutely right. I mean, when you think about high performance compute, and especially in the context of AI, and

I've said this many, many times before, AI is driven by massive amounts of data and about also understanding the correlation between those data elements and then overlaying that with smart software. But -- and I also believe actually what I'm seeing and what I'm hearing is that IoT in the industrial space will actually be an -- will be an area where we will see a lot of AI applications.

Well, in order to collect all that data, you need sensors, because you have all kinds of examples, whether it's the car or whether it's life science, medical equipment, it's about sensing. And that is really the domain of mainstream semiconductors. I don't like the word mature, like it's something which is old fashioned that you don't need it. It is mainstream and it's critical in the -- I would say in the -- I would call it amalgamation of mainstream semiconductors and I would call advanced semiconductors.

And this is also why you cannot distinguish the growth of one against the other. I mean, you need both. And I think what you will see is that the growth of the industry, and especially high-performance compute will be driven by the value that is going to be created. Yet, cost is a significant issue, but Moore's law is an empirical economics. And when you create more value of the transistor, the new transistor, the next generation transistor, then it costs, then you're going to grow.

And when you think about AI and with some of these examples and especially in the software space where you see productivity, just calculated productivity at voltages [ph] of 30% to 50%, then the value of the next generation transition will be huge. Now, if you ask me, Peter, what's going to be the next killer replication? I'm going to give you the same answer as I gave you for the last 25 years and I have no clue. But what I do know is that when the value of that transistor, next generation transistor, whether it's -- and it's particularly driven by high performance compute, its energy efficient performance, if we bring that to life and we'll bring it to life together with our customers because we have High NA and potentially Hyper NA, then the, I would say, amalgamated consolidated demand for also mainstream semiconductors will go up. So this is what I believe and this is what I see when I talk to customers and I talk to actually users of the value in the space. And this is why I'm so confident about the long-term future. And it's all connected, so you cannot distinguish the two.

Q - C.J. Muse {[BIO 6507553 <GO>](#)}

Very helpful. Thank you. And Roger, a follow up question on your backlog comment around the EUR4 billion plus required to hit the midpoint over the next three quarters. I guess as part of that, if you had to think about the higher end of the range, I think that would add two plus per quarter. And then also, if I look at your backlog today, excluding High NA, it's still sitting at roughly 18 months. And so obviously I would expect your backlog exiting '24 to have tools that will be shipped in '26. So is EUR4 billion the right number we should be thinking about for the next three quarters or should be it -- should it be significantly higher?

A - Roger Dassen {[BIO 15064806 <GO>](#)}

So if you're looking at the midpoint, it is EUR4 billion. And you shouldn't underestimate what we've already in that calculation taken out for the post '25 period. And I'm sure you're all going to do the math, right, because it's not all that difficult to do the math. And then you're probably going to figure out that that's a pretty healthy number and probably a number that maybe exceeds a little bit what you currently have in your

models in terms of High NA and that's okay. But that's a calculation that I leave up to you to make, but that has clearly been recognized in our calculation. So we've taken out whatever is for '25, and our focus of the EUR4 billion really is what pertains to 2025. And to your question, C.J., if you're looking at the high end of the range, there's a EUR5 billion delta last I saw between 35 and 40. And 5 divided by 3 gets you to 1.7, right? So that's it. So then there would be 1.7 needed more in order to get to the high end of the -- of that bandwidth at the beginning of the year.

A - Peter Wennink {BIO 1852674 <GO>}

(inaudible)

A - Roger Dassen {BIO 15064806 <GO>}

Exactly, yeah.

Q - C.J. Muse {BIO 6507553 <GO>}

Thank you.

A - Roger Dassen {BIO 15064806 <GO>}

Welcome.

Operator

Thank you. We'll now go to the next question. And your next question comes from the line of Alexander Duval from Goldman Sachs. Please go ahead.

Q - Alexander Duval {BIO 16682293 <GO>}

Hi, everyone. Many thanks, Peter, for everything over the years. I have one quick question and then a follow up. The first is on China services. We've seen a number of news articles talking about US government asking the Dutch government to prevent servicing of certain aspects of the Installed Base in China. I had a lot of investor questions on this and just wondered if you could provide any perspective on potential implications. Given especially that tools don't work without services, is it fair to assume any ban would likely not encompass all China services revenues?

And secondly, on electrification, you mentioned that electrification could be a potential driver of litho demand. We've seen a number of news articles talking about the need for technology to help deal with strong EUV driven power needs in the grid as well as AI server driven demand. So wondered if you could provide some context on what this means for longer term litho demand? Many thanks.

A - Peter Wennink {BIO 1852674 <GO>}

Yes, I think on the China services, yes, we're probably reading the same articles. So, yeah, that has been a discussion between the two governments and let it be a discussion between the two governments. I mean, of course, we're providing input. We're providing input of the size of the -- and the type of services and I think it's all being taken into consideration to determine in the end what the real problem is. And I think that is something which governments will need to discuss because probably it's all going to be in their discussion on what they call the national security interest. So we just provided

with the information and I think currently we have -- there's nothing that stops us servicing the Installed Base in China today.

On the electrification, I think we've said this many times when we talked about the growth of the industry and it also refers back to the question that C.J. asked. Yes, the grid electrification, the build of the grid, the investments in renewable energy and the investment in a smart grid will be a significant driver -- is a significant driver for mainstream semiconductors. So that is indeed true. Electrification, you think about the vehicle is also about mainstream semiconductors and advanced semiconductors.

So yes, I mean, this is not a requirement or a request of one part of an industry. It's all connected. So you are indeed correct. I think this is what people now start to realize that if we want to invest in renewable energy. Take the Netherlands for as a particular example, there's a significant investment in solar panels and in wind, which actually means that we need to upgrade the grid. And the grid needs to be smart because there's not going to be a constant supply of those electrons, so that needs to be managed. So, it's a complete overhaul of the electric grid needed.

Yeah, that's absolutely true and this is why we need more mainstream semiconductors. And we need a lot -- more of them. And Roger pointed out, we said at the capital market at 2022, 380,000 wafer starts per year need to be added. Well, we didn't do that last year. We were lower than that. So we're running behind our own model. So I'm happy that around the world investments in mainstream semiconductors take place because we need them.

A - Christophe Fouquet {[BIO 20524437 <GO>](#)}

And Alexander, it's everywhere. So it's in the generation of power, as Peter just said, solar panels, windmills, it's in the distribution, the net -- it's in the storage because that's another one, right. You need the electricity at a point in time where it's not being generated and it's in the usage, like the EUV. So it's everywhere. And therefore, this was a very significant part, as Peter just said, of the 380 that we talked about. Of course, we'll review this again when November comes, but I think the world is clearly appreciating a little bit better by now, how significant the electrification is and what it does for the demand for mature chips.

Q - Alexander Duval {[BIO 16682293 <GO>](#)}

Fascinating. Thank you very much.

Operator

Thank you. We will now go to the next question and your next question comes from the line of Chris Caso from Wolfe Research. Please go ahead.

Q - Chris Caso {[BIO 4815032 <GO>](#)}

Yes, thank you. Good morning. First question is regarding the guidance implied for the second half of this year, obviously expecting an acceleration in revenue. Can you give some detail on what you expect to lead that revenue growth in the second half? And specifically on China, you characterize that as strong right now. Do you expect that to remain strong as you go into the second half?

A - Roger Dassen {[BIO 15064806 <GO>](#)}

Yes. So the second half, I think this is very much in line with the orders that we have today because we're fully booked for the year, so the shipment plans that we have. And this is across the board. So this is for some of the fab openings and some of the ramps that I think all of you are aware that have been scheduled for 2024, both in logic and in memory. So this really is across the board.

As you know, there is - we're fairly conservative, I would say, on the Installed Base business, right. So the Installed Base business in the second half is only a little bit more than what we have in the first half. And I think that's still potential that I think there is for the year that we see an uptick in that number, particularly as it relates to the potential for upgrades in the second half of the year. But it really is very much, I would say, across the board.

Q - Chris Caso {[BIO 4815032 <GO>](#)}

And in terms of China?

A - Roger Dassen {[BIO 15064806 <GO>](#)}

In terms of China, I think we already said that we expect China to continue to be quite strong. Of course, it depends a little bit on the sales volume [ph] to the rest of the world. There are still some tools where we are supply constraints, so there the demand composition could be such that there are some limitations to what we can ship to China. But normally, as we said, the demand for China is very strong, so all the reasons to expect the strong sales into China to continue for the rest of the year as well.

A - Peter Wennink {[BIO 1852674 <GO>](#)}

And we are fully booked for 2024 and we are in the habit of actually shipping what we booked. So there's little doubt in our mind that 2024 will just turn out the way that we -- that we gave you as an outlook.

Q - Chris Caso {[BIO 4815032 <GO>](#)}

Understood. And as a follow up, follow up is on memory. And you spoke last quarter and the very strong bookings last quarter were a lot of technology buys on memory. Could you give us some sense on when you expect some of the follow through from perhaps some of the DPV orders that would supply capacity for memory? We've seen some green shoots in the memory market. Just interested what your customers are telling you regarding that capacity.

A - Roger Dassen {[BIO 15064806 <GO>](#)}

Well, it's very clear, as we mentioned before, that the utilization of memory is going up and we've seen that now quite sustainably for quite a while. And that -- if that is sustained as we expect, as the market expects, as our customers expect, then you should see in the second half of this year, you should see that memory is not just, let's say, technology transition, but it's also really the addition of bits. So that's I think very much in line with the current developments that we see in the market. So in the second half and one of the drivers why indeed you see this step up from the second half to the first half indeed is that you will see additional building of the capacity in the memory market.

A - Peter Wennink {BIO 1852674 <GO>}

Right.

Q - Chris Caso {BIO 4815032 <GO>}

From a -- and you're saying that from a revenue standpoint in addition to an order standpoint?

A - Roger Dassen {BIO 15064806 <GO>}

I say that from the revenue standpoint. Exactly right, yeah.

Q - Chris Caso {BIO 4815032 <GO>}

From revenue standpoint.

A - Roger Dassen {BIO 15064806 <GO>}

Yes.

Q - Chris Caso {BIO 4815032 <GO>}

Thank you.

Operator

Thank you. We will now go to the next question. And your next question comes from the line of Mehdi Hosseini from Susquehanna. Please go ahead.

Q - Mehdi Hosseini {BIO 4362002 <GO>}

Yes, thanks for taking my question. But before, Peter, want to wish you best of luck in your next endeavor and hopefully running to you at some launch -- at some April launch. And Christophe, hope to see you more often in US. Back to my question. Want to understand two things. First, what is the mix of backlog from attributed to China excluding EUV? And then second question, Roger, can you update us on the NXE:3800E? How is the throughput with the early system shipment in the first half and how is it going to improve in the second half?

A - Roger Dassen {BIO 15064806 <GO>}

Yes, so nice try, Mehdi, on China, but we're not disclosing that. We're not disclosing the composition of the backlog in a geographic sense. So we said before at some point that the China -- we said at once that China was a little bit over 20% in the backlog and that as a result of that, no one should be surprised that the sales is around that percentage. That's not dramatically changed. So I think we're still in that ballpark, but we're not going to be very specific on that. But at least that gives you a bit of a ballpark.

As we said on the introduction of the 3800, we said before that this will come in full configuration in a couple of months' time. So in the second half, we'll get it to 220 wafers per hour. Very quickly it'll get to 195. So it's several steps. So first, very rapidly, getting it to 195 wafers per hour. And then the twos that are going to leave the factory somewhere in the second half or early in the second half of this year will be 220 and the ones that

are still at 195 will get an upgrade, such that I would say early in next year, you'll see the entire fleet being at 220 wafers per hour. That's the plan, Mehdi.

Q - Mehdi Hosseini {BIO 4362002 <GO>}

To what extent that improvement towards 200 is a factor or driver in driving that EUR4 billion plus per quarter new orders?

A - Roger Dassen {BIO 15064806 <GO>}

In essence, it isn't. This is a revenue recognition thing, right? So we'll defer some of the revenue until the point in time where the ultimate configuration (inaudible) 220 is being achieved. So there is a revenue recognition issue here. It's not a PO issue because the customer signed a PO or the tool at full spec.

Q - Mehdi Hosseini {BIO 4362002 <GO>}

Got it. Thank you.

A - Roger Dassen {BIO 15064806 <GO>}

You're welcome.

A - Skip Miller {BIO 20244900 <GO>}

All right, so I think we have time for one last question. If you're unable to get through on this call and still have questions, please feel free to contact the ASML Investor Relations Department with your question. Now operator, may we have the last caller, please?

Operator

Thank you. We will now take the last question. And the question comes from the line of Didier Scemama from Bank of America. Please go ahead.

Q - Didier Scemama {BIO 21301064 <GO>}

Hey, good afternoon. Thank you for squeezing me in. And warm congratulations, Peter. And if Martin is listening, thank you so much for everything really. It's been an amazing privilege for me to cover the company for the last 24 years. And congratulations also to Christophe for the promotion to CEO job. I've got one question on EUV and DRAM, if I may. The question is really, first of all, can you tell us, roughly speaking, of your wafer capacity -- of the wafer capacity deployed in the industry. How much has transitioned to EUV yet? And how much do you think that could evolve to, as the industry adopt EUV for DDR5 and HBM?

And then related to that, I think in the past you had mentioned that over the longer term, I think 2030 was the sort of timeframe, the EUV unique breakdown could be 70-30 between logic and memory or DRAM. Do you think that the layer count increase we see with HBM could drive that to maybe a 50-50 sort of split over the course of the next five or 10 years? Thank you so much.

A - Christophe Fouquet {BIO 20524437 <GO>}

Hi, this is Christophe. Yes, I think your question about the exact capacity is quite difficult to answer. I don't think we have all the elements to do that. What I can share with you is the trend we see on DRAM, basically for EUV where our customer on map basically is calling for an increase node on node of the number of EUV layers. And I think we see that basically already for a few nodes. And that, I think, is with the current visibility of the roadmap we have from our customers. So we see basically the number of layers continue to increase to execute on the DRAM roadmap.

The high bandwidth memory doesn't change that dramatically because the device itself is very similar. What is changing is the ratio between the array and logic mostly, which typically called for larger die and therefore more wafer capacity needs. So there you have a translation in volume, but that's not related to the number of EUV layer, that's more related to the number of wafer you will need to do a high bandwidth memory.

And there again, I think Roger mentioned it before, we're still trying -- I think our customers are also still trying to understand exactly what would be the effect of high bandwidth memory on the overall capacity needs for DRAM in the next few months. But I think that the jury is still out on that and we are working very hard on our side to try to understand that with the idea that when we come to all of you in November 14, we understand that a bit more, but I think it's work in progress also on our customer side.

A - Peter Wennink {[BIO 1852674 <GO>](#)}

It's also the 70-30, the 70-30 mix.

A - Christophe Fouquet {[BIO 20524437 <GO>](#)}

Yes.

A - Peter Wennink {[BIO 1852674 <GO>](#)}

Which is also true for the 70-30 mix because it's all connected.

Q - Didier Scemama {[BIO 21301064 <GO>](#)}

Thank you so much.

A - Skip Miller {[BIO 20244900 <GO>](#)}

All right. As this will be the last quarterly call for Peter, I'd also like to take a moment to publicly state to Peter that on behalf of the ASML IR team, it's been our extreme pleasure to have worked with you over many years. Thank you for your leadership, so many great memories, and countless contributions to ASML. Congratulations on your retirement. We wish you and your family all the best.

A - Peter Wennink {[BIO 1852674 <GO>](#)}

I could have never done it without you guys. That's absolutely clear (inaudible).

A - Skip Miller {[BIO 20244900 <GO>](#)}

Now, on behalf of ASML, I'd like to thank you all for joining us today. Operator, if you could formally conclude the call, I'd appreciate it. Thank you.

Operator

Thank you. This concludes the ASML 2024 first quarter financial results conference call. Thank you for participating. You may now disconnect.

This transcript may not be 100 percent accurate and may contain misspellings and other inaccuracies. This transcript is provided "as is", without express or implied warranties of any kind. Bloomberg retains all rights to this transcript and provides it solely for your personal, non-commercial use. Bloomberg, its suppliers and third-party agents shall have no liability for errors in this transcript or for lost profits, losses, or direct, indirect, incidental, consequential, special or punitive damages in connection with the furnishing, performance or use of such transcript. Neither the information nor any opinion expressed in this transcript constitutes a solicitation of the purchase or sale of securities or commodities. Any opinion expressed in the transcript does not necessarily reflect the views of Bloomberg LP. © COPYRIGHT 2026, BLOOMBERG LP. All rights reserved. Any reproduction, redistribution or retransmission is expressly prohibited.